

Worksheet 3C: Quadratics — Real-World Applications

Annotated Revision Guide

AIML Foundations Mathematics Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

Part A: Projectile Motion (Problems 1–12)

Concept. Height follows $h(t) = h_0 + v_0t - 5t^2$ (using $g = 10 \text{ m/s}^2$). This is a **downward-opening parabola**. The vertex gives the **maximum height** at time $t^* = -b/(2a)$. Setting $h(t) = 0$ and solving the quadratic gives the **landing time**.

Method. (1) Identify $a = -5$, $b = v_0$, $c = h_0$. (2) Maximum at $t^* = v_0/10$. (3) Land when $h(t) = 0$: use the quadratic formula or factorise. (4) Keep only non-negative solutions (time runs forward).

Watch out. Two solutions when landing on the ground: one negative (before launch, discard), one positive (actual landing time). When thrown *downward*, $v_0 < 0$ appears in the formula.

Where this leads. Worksheet 3D puts these parabolas into vertex form $h = -5(t - t^*)^2 + h_{\text{max}}$. The derivative $h'(t) = v_0 - 10t$ gives instantaneous velocity; $h'(t) = 0$ at the peak.

Explore further. Galileo proved this parabolic trajectory in *Discorsi* (1638). Air resistance deforms the parabola for fast objects — see Trefethen & Trefethen, *The ... Football* (2005).

1. $h(t) = 20t - 5t^2$ (ground level, $v_0 = 20 \text{ m/s}$).

(a) $t = 1$: $h = 20 - 5 = 15 \text{ m}$.

(b) $t = 2$: $h = 40 - 20 = 20 \text{ m}$.

(c) $t = 4$: $h = 80 - 80 = 0 \text{ m}$. Back on the ground.

2. Factor and find zeros of $h(t) = 20t - 5t^2$.

Factor. $h = 5t(4 - t)$.

Zeros. $t = 0$ (launch) and $t = 4 \text{ s}$ (landing).

3. Dropped from 45 m: $h(t) = 45 - 5t^2$.

(a) $t = 1$: $45 - 5 = 40 \text{ m}$.

(b) $t = 2$: $45 - 20 = 25 \text{ m}$.

(c) **Land when** $h = 0$: $5t^2 = 45 \Rightarrow t^2 = 9 \Rightarrow t = 3 \text{ s}$.

4. Thrown down from 80 m cliff: $h(t) = 80 - 10t - 5t^2$.

- (a) **Negative v_0 :** Downward initial velocity means height decreases immediately.
- (b) **Land:** $5t^2 + 10t - 80 = 0 \Rightarrow t^2 + 2t - 16 = 0 \Rightarrow t = \frac{-2 \pm \sqrt{4+64}}{2} = \frac{-2 \pm \sqrt{68}}{2} \approx \frac{-2 \pm 8.25}{2} \approx 3.12$ s (using positive root).
- (c) **Speed:** Speeding up — gravity continuously accelerates the ball downward.
- Note: Exact: $t = -1 + \sqrt{17} \approx 3.12$ s.*

5. Rocket from 20 m platform: $h(t) = -5t^2 + 50t + 20$.

- (a) **Standard form:** Already in standard form $-5t^2 + 50t + 20$.
- (b) $h(0) = 20$ m. Matches the 20 m platform. ✓
- (c) **Quad formula:** $a = -5, b = 50, c = 20$. Discriminant $= 2500 + 400 = 2900$.
 $t = \frac{-50 \pm \sqrt{2900}}{-10} = \frac{-50 \pm 53.85}{-10}$.
- Solutions:** $t = \frac{-50+53.85}{-10} \approx -0.39$ s (discard) or $t = \frac{-50-53.85}{-10} \approx 10.39$ s.
- Note: (d) Two solutions because the parabola crosses $h = 0$ twice, but only once in the future.*

6. Maximum of $h(t) = -5t^2 + 20t$.

- (a) $a = -5, b = 20$.
- (b) $t^* = -b/(2a)$. $t^* = -20/(2 \times -5) = -20/(-10) = 2$ s.
- (c) **Max height.** $h(2) = -5(4) + 20(2) = -20 + 40 = 20$ m.

7. Rocket $h = -5t^2 + 50t + 20$: max height.

- (a) **Time.** $t^* = -50/(2 \times -5) = 5$ s.
- (b) **Height.** $h(5) = -5(25) + 250 + 20 = -125 + 270 = 145$ m.

8. Football: $h = -5t^2 + 15t + 1$.

- (a) **Initial height.** $h(0) = 1$ m (kicker's foot height).
- (b) **Peak time.** $t^* = 15/10 = 1.5$ s.
- (c) **Max height.** $h(1.5) = -5(2.25) + 22.5 + 1 = -11.25 + 23.5 = 12.25$ m.
- (d) **Lands:** $-5t^2 + 15t + 1 = 0$; disc $= 225 + 20 = 245$; $t = (15 \pm \sqrt{245})/10 \approx (15 \pm 15.65)/10 \approx 3.06$ s.

9. Ball kicked: $v_x = 10, v_y = 15$ m/s. Trajectory $y(t) = 15t - 5t^2$.

- (a) $y(t)$. $y = 15t - 5t^2 = 5t(3 - t)$.
- (b) **Lands.** $y = 0 \Rightarrow t = 0$ (launch) or $t = 3$ s.
- (c) **Horizontal range.** $x = 10 \times 3 = 30$ m.

10. Eliminate t to get y as a function of x .

From $x = 10t$: $t = x/10$.

Substitute: $y = 15 \cdot \frac{x}{10} - 5 \cdot \frac{x^2}{100} = \frac{3x}{2} - \frac{x^2}{20}$.

Note: This is a parabola in x : horizontal axis is range, vertical is height. Same shape as the t -parabola, just stretched/scaled differently.

11. Ball from 2 m: $h = 2 + 12t - 5t^2$. When is $h = 10$ m?

Set up. $2 + 12t - 5t^2 = 10 \Rightarrow -5t^2 + 12t - 8 = 0 \Rightarrow 5t^2 - 12t + 8 = 0$.

Discriminant. $144 - 160 = -16 < 0$. Hmm — wait.

Recalculate. $-5t^2 + 12t + 2 - 10 = 0 \Rightarrow -5t^2 + 12t - 8 = 0$. Disc = $144 - 160 = -16$.

Note: The discriminant is negative, meaning the ball never reaches 10 m. Maximum height: $t^ = 12/10 = 1.2$ s; $h(1.2) = 2 + 14.4 - 7.2 = 9.2$ m < 10 m. So 10 m is above the maximum — the ball cannot reach it. This is a plausible exam catch.*

12. When are ball A ($h_A = 30t - 5t^2$) and ball B ($h_B = 25 + 10t - 5t^2$) at the same height?

Set equal. $30t - 5t^2 = 25 + 10t - 5t^2$.

Simplify. $30t = 25 + 10t \Rightarrow 20t = 25 \Rightarrow t = 1.25$ s.

Part B: Area Problems (Problems 13–20)

Concept. Geometric area problems often lead to quadratic equations. The strategy: let one dimension be w (or another variable), express everything else in terms of w , write the area formula, set equal to the given area, and solve.

Method. For rectangles: Area = length $\times$ width. For fencing: use the perimeter constraint to eliminate one variable. The maximum area under a fixed perimeter is found at the vertex of the resulting quadratic.

Watch out. Dimensions must be positive. Always check that your solutions give positive width and length. Discard any negative solution.

Where this leads. The fencing problem (14) is a classic optimisation problem. Calculus solves it in one step: $\frac{dA}{dw} = 0$. Here the vertex formula $w^* = -b/(2a)$ gives the same answer.

Explore further. The isoperimetric inequality says: among all shapes with a fixed perimeter, the *circle* encloses maximum area. The rectangle optimum is a special case.

13. Rectangle: length = $w + 3$, area = 70 m^2 .

(a) **Length.** $w + 3$.

(b) **Area.** $w(w + 3)$.

(c) **Solve.** $w^2 + 3w = 70 \Rightarrow w^2 + 3w - 70 = 0 \Rightarrow (w + 10)(w - 7) = 0 \Rightarrow w = 7$ m, length = 10 m.

14. Fencing: 100 m against a river (no fence on river side).

- (a) **Length.** $L = 100 - 2w$.
(b) **Area.** $A = w(100 - 2w) = -2w^2 + 100w$.
(c) **Maximise.** $w^* = -100/(2 \times -2) = 25$ m.
(d) **Max area.** $A(25) = 25 \times 50 = 1250$ m².

15. Square: side increased by 4 cm, new area = 81 cm².

- (a) **Equation.** $(s + 4)^2 = 81$.
(b) **Solve.** $s + 4 = 9 \Rightarrow s = 5$ cm.

16. Product of two consecutive positive integers = 182.

- Equation.** $n(n + 1) = 182 \Rightarrow n^2 + n - 182 = 0$.
Factor. $(n + 14)(n - 13) = 0 \Rightarrow n = 13$. Integers: 13 and 14.

17. Product of two consecutive even integers = 288.

- Equation.** $n(n + 2) = 288 \Rightarrow n^2 + 2n - 288 = 0$.
Discriminant. $4 + 1152 = 1156 = 34^2$.
Solve. $n = \frac{-2+34}{2} = 16$. Integers: 16 and 18.

18. Photo (w wide, $w + 4$ long) in 2 cm frame, total area = 165 cm².

- (a) **Outer dimensions.** $(w + 4) \times (w + 8)$.
(b) **Area equation.** $(w + 4)(w + 8) = 165$.
(c) **Expand.** $w^2 + 12w + 32 = 165 \Rightarrow w^2 + 12w - 133 = 0 \Rightarrow (w + 19)(w - 7) = 0 \Rightarrow w = 7$ cm, length = 11 cm.

19. Box from 20×30 sheet, corner squares of side x .

- (a) **Box dimensions.** $(20 - 2x) \times (30 - 2x) \times x$.
(b) **Volume.** $V = x(20 - 2x)(30 - 2x)$.
(c) $V = 1008$: $x(20 - 2x)(30 - 2x) = 1008$. Try $x = 3$: $3 \times 14 \times 24 = 1008$. ✓
 $x = 3$ cm.

20. Sum of a number and its square = 72.

Equation. $n + n^2 = 72 \Rightarrow n^2 + n - 72 = 0 \Rightarrow (n + 9)(n - 8) = 0$.

Solutions. $n = 8$ or $n = -9$. (Both valid as "a number".).

Part C: Taylor Polynomial Approximations (Problems 21–30)

Concept. Transcendental functions like $\sin$, $\cos$, e^x can be *approximated* by polynomials near $x = 0$: $\sin \theta \approx \theta - \frac{\theta^3}{6}$, $\cos \theta \approx 1 - \frac{\theta^2}{2}$, $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$. These are **Taylor/Maclaurin series** truncated to a few terms.

Method. Substitute the given value into the polynomial and evaluate. Compare to the calculator value to see how accurate the approximation is. Accuracy improves as (a) more terms are included or (b) $|\theta|$ is smaller.

Watch out. These approximations work well only for *small* arguments (near 0). For $\theta = 1$ radian, $\sin \theta \approx \theta - \theta^3/6 = 1 - 1/6 \approx 0.833$, while true $\sin(1) \approx 0.841$ — reasonable but not perfect. For $\theta = 3$, the approximation is very poor.

Where this leads. The full Taylor series converges everywhere for e^x , $\sin x$, $\cos x$. Truncating gives a polynomial approximation useful for numerical algorithms. Problem 30 previews **Euler's formula** $e^{i\theta} = \cos \theta + i \sin \theta$.

Explore further. Strang, *Calculus* (free online, MIT OpenCourseWare), Ch. 10. Also: Maor, *e: The Story of a Number* (1994).

21. $\sin \theta \approx \theta - \theta^3/6$ at $\theta = 0.1$.

Calculate. $0.1 - \frac{0.001}{6} = 0.1 - 0.0001\bar{6} \approx 0.09983$.

True value. $\sin(0.1) = 0.09983$. Extremely close!

22. $\cos \theta \approx 1 - \theta^2/2$ at $\theta = 0.2$.

Calculate. $1 - 0.04/2 = 1 - 0.02 = 0.98$.

True value. $\cos(0.2) = 0.9801$. Error ≈ 0.0001 .

23. Factor θ out of $\sin \theta \approx \theta - \theta^3/6$.

Factor. $\theta(1 - \theta^2/6)$.

Small θ . When $\theta \approx 0$: $\theta^2/6 \approx 0$, so $\sin \theta \approx \theta$. This is the small-angle approximation.

24. Cosine approximation $1 - \theta^2/2$ as a parabola.

(a) **Vertex.** At $\theta = 0$: $1 - 0 = 1$. Vertex is $(0, 1)$.

(b) **Matches cos.** $\cos(0) = 1$. ✓ The true cosine also has a maximum of 1 at $\theta = 0$.

25. When is $\sin \theta \approx \theta$ accurate?

- (a) **Range.** When $\theta^2/6$ is negligible, i.e. $\theta \lesssim 0.1$ rad (about 5.7°).
 (b) **1% accuracy.** Need $|1 - \cos \theta| < 0.01$, i.e. $\theta^2/2 < 0.01 \Rightarrow \theta < \sqrt{0.02} \approx 0.14$ rad.

26. $\tan \theta \approx \theta + \theta^3/3$ at $\theta = 0.1$.

- (a) **Factor.** $\theta(1 + \theta^2/3)$.
 (b) **Calculate.** $0.1(1 + 0.01/3) \approx 0.1 \times 1.00333 \approx 0.10033$.
 (c) **True value.** $\tan(0.1) \approx 0.10033$. ✓

27. Check $\sin^2 \theta + \cos^2 \theta \approx 1$ at $\theta = 0.1$.

$\sin(0.1) \approx 0.09983$. $\sin^2 \approx 0.009967$.
 $\cos(0.1) \approx 0.99500$. $\cos^2 \approx 0.99003$.
Sum. $0.009967 + 0.990033 = 1.000$. ✓

28. $e^x \approx 1 + x + x^2/2$ at $x = 0.1$.

Calculate. $1 + 0.1 + 0.005 = 1.105$.
True value. $e^{0.1} = 1.10517$. Error ≈ 0.00017 .

29. $e^x \approx 1 + x + x^2/2 + x^3/6$ at $x = 0.5$.

- (a) **Degree.** 3.
 (b) **Calculate.** $1 + 0.5 + 0.125 + 0.0208\bar{3} = 1.6458$.
 (c) **True value.** $e^{0.5} = 1.6487$. Error ≈ 0.003 . Very good for degree 3.

30. Euler's formula consistency check.

$e^{i\theta}$ **expansion.** $\approx 1 + i\theta + \frac{(i\theta)^2}{2} = 1 + i\theta - \frac{\theta^2}{2}$ (since $i^2 = -1$).
 $\cos \theta + i \sin \theta \approx (1 - \frac{\theta^2}{2}) + i\theta$.
Identical? Yes! Real parts match $(1 - \theta^2/2)$ and imaginary parts match $(i\theta)$. ✓
Note: This is a polynomial-level verification of Euler's profound formula $e^{i\pi} + 1 = 0$.

Part D: Binomial Expansion and Probability (Problems 31–40)

Concept. $(H + T)^n$ expanded gives all possible outcomes of n coin flips. The coefficient of $H^k T^{n-k}$ is $\binom{n}{k}$, the number of ways to get exactly k heads. For a biased coin with $P(H) = p$, replace H by p and T by $1 - p$.

Method. Expand using Pascal's Triangle or the binomial theorem. Each coefficient

$\binom{n}{k}$ can be read from row n of Pascal's Triangle or computed as $\frac{n!}{k!(n-k)!}$.

Watch out. The exponents in $(p+q)^n$ represent probabilities, not counts. Always verify: the sum of all terms $= (p+q)^n = 1^n = 1$ for $p+q=1$.

Where this leads. The binomial distribution $B(n, p)$ is the foundation of statistical tests, A/B testing, and the normal approximation via the Central Limit Theorem.

Explore further. Feller, *An Introduction to Probability Theory* (Vol. 1, 1968) — the classic, still unsurpassed.

31. Expand $(H+T)^3$.

Result. $H^3 + 3H^2T + 3HT^2 + T^3$.

32. From $(H+T)^3$, how many ways to get each outcome?

0 heads. 1 way: T^3 .

1 head. 3 ways: HTT, THT, TTH .

2 heads. 3 ways.

3 heads. 1 way: H^3 .

Total. $1 + 3 + 3 + 1 = 8 = 2^3$. ✓

33. Fair coin: expand $(0.5 + 0.5)^3$.

Probabilities. 0 heads: 0.125; 1 head: 0.375; 2 heads: 0.375; 3 heads: 0.125.

34. Biased coin $p = 0.6$: expand $(0.6 + 0.4)^3$.

(a) **0 heads.** $0.4^3 = 0.064$.

(b) **1 head.** $3 \times 0.6 \times 0.16 = 0.288$.

(c) **2 heads.** $3 \times 0.36 \times 0.4 = 0.432$.

(d) **3 heads.** $0.6^3 = 0.216$. Sum = 1.000. ✓

35. 4 fair coin flips (row 4 of Pascal's: 1, 4, 6, 4, 1).

(a) **Exactly 2 heads.** $\binom{4}{2}(0.5)^4 = 6/16 = 0.375$.

(b) **At least 3 heads.** $P(3) + P(4) = (4 + 1)/16 = 5/16 = 0.3125$.

36. Expand $(p+q)^4$ where $p+q=1$.

Result. $p^4 + 4p^3q + 6p^2q^2 + 4pq^3 + q^4$.

37. 5-question MCQ, $p = 0.25$: probability of exactly 3 correct.

Row 5 of Pascal's: 1, 5, 10, 10, 5, 1.

$P(X = 3)$: $\binom{5}{3}(0.25)^3(0.75)^2 = 10 \times 0.015625 \times 0.5625 \approx 0.0879$.

38. Same test: probability of at least 4 correct by guessing.

$P(X = 4)$. $5 \times (0.25)^4 \times 0.75 = 5 \times 0.00391 \times 0.75 \approx 0.01465$.

$P(X = 5)$. $(0.25)^5 \approx 0.000977$.

Total. $\approx 0.01563 \approx 1.6\%$.

39. In $(H + T)^4$, which term has the largest coefficient?

Term. $6H^2T^2$: coefficient 6. This is $\binom{4}{2} = 6$, the middle term.

Meaning. Getting exactly 2 heads out of 4 flips is the most likely single outcome.

40. Basketball: $p = 0.7$, 3 free throws.

(a) **All 3.** $0.7^3 = 0.343$.

(b) **At least 2.** $P(2) + P(3) = 3(0.7)^2(0.3) + (0.7)^3 = 3 \times 0.147 + 0.343 = 0.441 + 0.343 = 0.784$.

Part E: Compound Growth and Finance (Problems 41–50)

Concept. The compound interest formula $A = P(1 + r/n)^{nt}$ uses binomial expansion when n is small. Key special cases: annual compounding ($n = 1$), continuous compounding ($n \rightarrow \infty$, gives Pe^{rt}), and depreciation (replace r with $-r$).

Method. For expansion: $(1 + r)^2 = 1 + 2r + r^2$; $(1 + r)^3 = 1 + 3r + 3r^2 + r^3$. For the Rule of 72: doubling time $\approx 72/(\text{interest rate } \%)$. Continuous compounding limit: $(1 + 1/n)^n \rightarrow e$.

Watch out. The approximation $(1 + r)^n \approx 1 + nr$ is only valid for small r and n . At $r = 0.1$, $n = 2$: exact = 1.21; approximate = 1.20. Fine. At $r = 0.5$, $n = 4$: exact = 5.0625; approximate = 3. Very wrong.

Where this leads. Continuous compounding Pe^{rt} is the solution to the differential equation $\frac{dA}{dt} = rA$: the growth rate is proportional to the current amount. This ODE appears throughout physics, chemistry, and biology.

Explore further. Bernoulli discovered $(1 + 1/n)^n \rightarrow e$ around 1683 while studying continuous compounding. Napier invented logarithms in 1614 partly to simplify

compound interest calculations.

41. €1000 at 5% for 2 years: $A = 1000(1.05)^2$.

(a) **Expand.** $(1 + 0.05)^2 = 1 + 0.10 + 0.0025 = 1.1025$.

(b) **Amount.** $1000 \times 1.1025 = \text{£ } 1102.50$.

(c) **Interest.** €102.50.

42. Same at 3 years: $(1.05)^3$.

Expand. $(1.05)^3 = 1 + 3(0.05) + 3(0.05)^2 + (0.05)^3 = 1 + 0.15 + 0.0075 + 0.000125 = 1.157625$.

Amount. $\approx \text{£ } 1157.63$.

43. Rule of 72: doubling time at 6%.

(a) **Estimate.** $72/6 = 12$ years.

(b) **Verify.** $(1.06)^{12} \approx 2.012$. ✓

44. $(1 + r)^2$ symbolically; evaluate at $r = 0.1$.

Expand. $1 + 2r + r^2$.

At $r = 0.1$. $1 + 0.2 + 0.01 = 1.21$. Growth factor of 21% over 2 years.

45. Approximation $(1 + r)^n \approx 1 + nr$.

(a) $(1.02)^5 \approx 1 + 5(0.02) = 1.10$.

(b) **Exact.** $(1.02)^5 = 1.10408$.

(c) **Good when.** r is small and n is small; the quadratic and higher terms are negligible.

46. Inflation 3%: value of €100 after n years $= 100(0.97)^n$.

(a) $n = 2$. $100 \times 0.9409 = \text{£ } 94.09$.

(b) $n = 5$. $100 \times 0.97^5 \approx 100 \times 0.859 = \text{£ } 85.87$.

(c) **Halved when.** $(0.97)^n = 0.5 \Rightarrow n = \ln(0.5)/\ln(0.97) \approx 22.8$ years.

47. Population: $P(t) = 50000(1.02)^t$.

- (a) $t = 3$. $50000 \times 1.02^3 = 50000 \times 1.0612 \approx 53060$.
 (b) **Reach 60000.** $(1.02)^t = 1.2 \Rightarrow t = \ln(1.2)/\ln(1.02) \approx 9.5$ years.

48. Car depreciates 15%/yr: $V(t) = 20000(0.85)^t$.

- (a) $t = 1$. €17000.
 (b) $t = 3$. $20000 \times 0.85^3 = 20000 \times 0.6141 \approx \text{£} 12282$.
 (c) $< \text{£}10000$. $(0.85)^t < 0.5 \Rightarrow t > \ln(0.5)/\ln(0.85) \approx 4.3$ years.

49. Approach to e : $(1 + 1/n)^n$.

- $n = 1$. 2.
 $n = 2$. 2.25.
 $n = 10$. ≈ 2.594 .
 $n = 100$. ≈ 2.705 .
 $n \rightarrow \infty$. $\rightarrow e \approx 2.71828$.

50. Loan payment: €10000 at $r = 0.005$ /month for 24 months.

- (a) $(1.005)^{24} \approx 1.1272$.
 (b) **Monthly payment.** $10000 \times \frac{0.005 \times 1.1272}{1.1272 - 1} = 10000 \times \frac{0.005636}{0.1272} \approx \text{£} 443$.
 (c) **Total paid.** $443 \times 24 \approx \text{£} 10632$.
 (d) **Interest.** $\approx \text{£} 632$.

Note: The exact payment formula gives €443.21, for a total of €10,637 and €637 in interest.